

LP2(P1 BFS DFS)

Implement depth first search algorithm and Breadth First Search algorithm. Use an undirected graph and develop a recursive algorithm for searching all the vertices of a graph or tree data structure.

```
from collections import deque
```

```
def dfs(v, g, n):
```

```
    if n not in v:
```

```
        print(n, end=" ")
```

```
        v.add(n)
```

```
        for i in g[n]:
```

```
            dfs(v, g, i)
```

```
def bfs(v, g, n):
```

```
    q = deque([n])
```

```
    v.add(n)
```

```
    while q:
```

```
        s = q.popleft()
```

```
        print(s, end=" ")
```

```
        for i in g[s]:
```

```
            if i not in v:
```

```
                v.add(i)
```

```
                q.append(i)
```

```
g = {}  
  
n = int(input("Enter number of nodes: "))  
  
for i in range(1, n+1):  
    g[i] = list(map(int, input(f"Enter neighbors of {i}: ").split()))  
  
print("\nDFS:")  
dfs(set(), g, 1)  
  
print("\nBFS:")  
bfs(set(), g, 1)
```

DFS & BFS – Complete Viva Question Bank

★ Basic Theory

1. What is a graph?

A graph is a collection of vertices (nodes) and edges (connections between nodes).

2. What are vertices and edges?

- Vertices → nodes
 - Edges → connections between nodes
-

3. What is graph traversal?

Visiting all nodes of a graph in a systematic way.

4. What are the types of graph traversal?

- DFS (Depth First Search)
 - BFS (Breadth First Search)
-

★ DFS (Depth First Search)

5. What is DFS?

DFS explores a graph by going as deep as possible before backtracking.

6. Which data structure is used in DFS?

Stack (or recursion).

7. Why recursion is used in DFS?

Because recursion internally uses a stack.

8. What is the time complexity of DFS?

$O(V + E)$

9. What are applications of DFS?

- Maze solving
 - Topological sorting
 - Cycle detection
 - Path finding
-

10. What is backtracking in DFS?

Returning to previous node after exploring a path fully.

☆ BFS (Breadth First Search)

11. What is BFS?

BFS visits nodes level by level starting from the source node.

12. Which data structure is used in BFS?

Queue (FIFO).

13. Why queue is used in BFS?

Because it processes nodes in order of arrival.

14. What is time complexity of BFS?

$O(V + E)$

15. Applications of BFS?

- Shortest path in unweighted graph
 - Social networks
 - Web crawling
-

☆ DFS vs BFS

16. Difference between DFS and BFS?

DFS	BFS
Uses stack	Uses queue
Goes deep	Goes level-wise
Uses less memory	Uses more memory
May not give shortest path	Gives shortest path

17. Which gives shortest path?

BFS

18. Which is faster?

Depends on problem, but BFS is better for shortest path.

☆ Graph Input / Code Related

19. Why do we use adjacency list?

Because it saves memory and is efficient for sparse graphs.

20. What is visited set?

It keeps track of visited nodes to avoid loops.

21. Why is visited set required?

To prevent infinite recursion or repeated nodes.

22. What happens if visited is not used?

Infinite loop in cyclic graphs.

23. What is recursion base condition in DFS?

When node is already visited.

☆ **Practical / Code Questions**

24. What input format is used?

User inputs nodes and their neighbors.

25. Why do we use graph[node]?

To access adjacent nodes.

26. What is the starting node?

The node from which traversal begins (usually 1 or user input).

27. What is output of DFS and BFS?

Order in which nodes are visited.

28. What is undirected graph?

Graph where edges go both ways.

☆ **Advanced Viva Questions**

29. Can DFS get stuck?

Yes, if visited set is not used.

30. Can BFS get stuck?

Yes, without visited tracking.

31. Which is better for large graphs?

Depends:

- DFS → less memory
 - BFS → better for shortest path
-

32. Is BFS always optimal?

Yes for unweighted graphs.

33. Is DFS optimal?

No, not always.

☆ One-line Important Questions

34. Define DFS in one line:

DFS explores as deep as possible before backtracking.

35. Define BFS in one line:

BFS explores nodes level by level using a queue.

36. What is graph traversal?

Visiting all nodes of a graph systematically.

🔥 MOST IMPORTANT (EXAM FAVORITE)

37. Real-life applications of DFS & BFS?

- Google Maps
- Social networks
- AI pathfinding
- Web crawling
- Games